Quinn Asena

Data Scientist & Ecologist

Cary Institute of Ecosystem Studies

Forest Futures Lab, Millbrook, NY

CONTACT

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LAB AFFILIATIONS

Forest Futures Lab
Cary Institute

Williams Paleoecology
UW-Madison

Perry Lab
U. Auckland

Neotoma Paleoecology DB
Leadership Council

SKILLS & TOOLS

R · Python · C++ · Bash · Slurm/ PBS · HPC · Git · SQL · Apache Arrow

INTERESTS

Classical piano · Rock climbing · Art

RESEARCH INTERESTS

I am an **ecologist and data scientist** focused on how ecosystems change over time and space, and on building the data infrastructure needed to study them. My current work includes deep learning methods for **wildfire risk mapping**, and process-based **landscape simulation** with CMIP6 climate scenarios on HPC systems to project forest biodiversity change. My background spans palaeoecology, contemporary ecosystem modelling, and the statistical and computational methods that link the two.

PROFESSIONAL EXPERIENCE

Data Scientist — *Millbrook, NY*
Cary Institute of Ecosystem Studies — *2025 – Present*

- Automated HPC workflows for process-based landscape simulations across boreal North America under multiple CMIP6 climate scenarios.
- Large-scale data management using SQL and Apache Arrow; integration of climate, soil, and elevation data from distributed archives (NASA DAACs, STAC).
- Applying deep learning models to project wildfire risk across boreal North America.

Postdoctoral Research Associate — *Madison, WI*
University of Wisconsin-Madison — *2022 – 2024*

- NSF-funded project developing state-space modelling methods to analyse palaeoecological records.
- Abrupt change in ecosystems project led by Jack Williams and Anthony Ives.

Engagement Specialist — *Auckland, NZ*
Centre for eResearch - University of Auckland — *2021 – 2022*

- Delivered seminars on data management, FAIR and CARE principles, and reproducible workflows; provided research-computing support for graduate students and research staff (virtual machines, data storage, analytical workflows).
- Technical skills: Linux, Bash, Slurm, HPC, and version control.

Data Analyst — *Auckland, NZ*
Auckland Council — *2021*

- Analysed citizen science bird count data and produced a technical report.

RESEARCH EXPERIENCE

Research Assistant — *Auckland, NZ*
University of Auckland — *2021 – 2022*

- Conducted a bibliometric analysis on the topic of climate justice.

Research Assistant — *York, UK*
Stockholm Environment Institute — *2016 – 2017*

- Reconstruction of fire regimes in the North York Moors.
- Microscope analysis and core preparation for charcoal and spheroidal carbonaceous particles.

TEACHING EXPERIENCE

PhD Student Mentor *Virtual*
SUNY *2025*

- One-on-one mentoring in statistics for palaeoecology.

Graduate Teaching Assistant & Assistant Coordinator *Auckland, NZ*
University of Auckland *2018 – 2020*

- Assistant coordinator for Discovering Environmental Modelling.
- Guest lecturer on population growth. Developed a Shiny application as a lecture aid.
- GTA for: Natural and Human Environmental Systems; Discovering Environmental Modelling; Environmental Science and Management.

EDUCATION

PhD, University of Auckland *Auckland, NZ*
School of Environment *2017 – 2021*

- Explored virtual ecological methods for generating pseudoproxy data to assess statistical inferences under data uncertainty.
- Supervised by George Perry and Janet Wilmshurst.

MEnv. Environmental Science *York, UK*
University of York *2012 – 2016*

- First-class integrated master's degree in environmental science. Master's research involved environmental monitoring using sensor networks and robotics.
- Courses included: Ecological Principles; Biogeography; Applied Ecology and Environmental Management; Climate Change: Science, Observation & Impacts; Research Skills and Statistical Methods.

Geocomputation & Machine Learning for Environmental Applications *Matera, Italy*
Basilicata University *2025*

- Certified course in machine learning for spatial analyses.

Certified Carpentries Instructor *Global (Virtual)*
The Carpentries *2023*

- Certified to teach and host Carpentries workshops in data science and computational skills.

Performance Composition & Theory *London, UK*
Royal Academy of Music *2005 – 2010*

- Study in performance, composition, and musical theory. Multiple competition awards.
- ABRSM Grade 8 Piano Performance.
- ABRSM Grade 5 Music Theory.

PUBLICATIONS

- Perrotti, A.G., Williams, J.W., **Asena, Q.**, Schlenker, N., Lam, J., Siedelmann, M., Russell, J.M., Ives, A.R. (In review). *Past novel ecosystems in Florida driven by fluctuating climate, fire, and megaherbivory.*
- Johnson, J.W., Ray-Cozzens, T., Shuman, B.N., Williams, J.W., Ives, A.R., **Asena, Q.**, Stefanova, V., Nelson, D.M. (In review). *Differential responses of American Beech and Eastern Hemlock to Holocene drought and warming: Insights from pollen $\delta^{13}C$ and multiproxy reconstructions.*

- **Asena, Q.**, Williams, J.W., Johnson, J.W., Shuman, B.N., Stefanova, V., Ives, A.R. (2026). Statistical analyses of ecological multinomial time series to identify environmental drivers and biotic interactions. *Methods in Ecology and Evolution.*
- **Asena, Q.**, Perry, G.L., Wilmshurst, J.M. (2026). Information loss in palaeoecological data from process and observer error. *Climate of the Past.*
- Lee, F., Bellvé, A.M., Alena, H., **Asena, Q.**, Perry, G.L. (2025). Using population viability analysis and fossil records to inform the conservation of pateke (*Anas chlorotis*). *New Zealand Journal of Ecology.*
- **Asena, Q.**, Perry, G.L., Wilmshurst, J.M. (2024). Is the past recoverable from the data? Pseudoproxy modelling of uncertainties in palaeoecological data. *The Holocene.*
- Parsons, M., **Asena, Q.**, Johnson, D., Nalau, J. (2024). A Bibliometric Review and Topic Analysis of Climate Justice Literature: Mapping Trends, Voices, and the Way Forward. *Climate Risk Management.* Code repository publicly available.
- Courtney-Mustaphi, C.J., Brahney, J., Aquino-López, M.A., Goring, S., Orton, K., Noronha, A., Czaplewski, J., **Asena, Q.**, Paton, S. (2019). Guidelines for Reporting and Archiving ^{210}Pb Sediment Chronologies to Improve Fidelity and Extend Data Lifecycle. *Quaternary Geochronology.*
- Heinemeyer, A., Burn, W.L., **Asena, Q.**, Jones, A., Ashby, M. (2019). Response to comment on Peatland carbon stocks and burn history. *GEO: Geography and Environment.*
- Heinemeyer, A., **Asena, Q.**, Burn, W.L., Jones, A. (2018). Peatland carbon stocks and burn history: Blanket bog peat core evidence highlights charcoal impacts on peat physical properties and long-term carbon storage. *GEO: Geography and Environment.*
- Simpkins, C.E., Lee, F., Powers, B.F., Anderson, S., **Asena, Q.**, Brock, J.M.R., Perry, G.L. (2018). Population viability analyses in New Zealand: a review. *New Zealand Journal of Ecology.*
- Hamilton, R., Brussel, T., **Asena, Q.**, Bruel, R., Marcisz, K., Słowiński, M., Morris, J. (2018). Assessing the links between resilience, disturbance, and functional traits in paleoecological datasets. *Past Global Changes Magazine, vol. 26(2), 87.*

SOFTWARE & OPEN SCIENCE

Palaeoecological data science: a data-driven curriculum *Curriculum*

- an open-source book for teaching palaeoecological data science.

multinomialTS *R package*

- R package for state-space modelling of multinomially distributed ecological time-series data.

State-Space Methods for Palaeoecological Hypothesis Testing *Workshop materials*

- Open workshop materials for state-space modelling of community and palaeoecological time series.

Authoring Collaborative Research Projects in Quarto *Workshop materials*

- Resources for collaborative research workflows using Git, GitHub, and Quarto (ResBaz 2022–2024).

fisheR *R package*

- R package for calculating Fisher's Information on ecological time-series data.

Population Growth Explorer *Shiny app*

- Interactive app exploring population growth equations for undergraduate teaching.

CONFERENCES, TALKS & WORKSHOPS

Methods in Ecology and Evolution *2026*
Virtual

- Talk: multinomialTS: a state-space approach for modelling community time series
- Materials: Talk and workshop resources

ESA 2026 - Ecological Society of America *2026*
Salt Lake City, UT

- **Speaker**: Assessing climate-driven compositional shifts in boreal forests using iLand

Neotoma Outreach *2026*
Virtual

- Talk: A virtual ecological approach to modelling uncertainties in palaeoecology
- Workshop: Hands-on workshop for age-depth modelling using Bchron

ESA 2025 - Ecological Society of America *2025*
Baltimore, MD

- **Speaker**: Modelling palaeoecological datasets using a state-space approach: two case studies

AGU - American Geophysical Union *2025*
New Orleans, LA

- Poster: Assessing climate-driven compositional shifts in boreal forests using iLand.

ESA 2024 - Ecological Society of America *2024*
Long Beach, CA

- **Speaker**: Hypothesis testing of the drivers of long-term ecological change in Michigan.
- **Workshop host**: State-space methods for testing hypotheses about environmental drivers and biotic interactions

INQUA 2023 - International Union for Quaternary Research *2023*
Rome, Italy

- **Lead Convenor**: Data science and palaeoecology: current intersections and advances

ESA 2023 - Ecological Society of America *2023*
Portland, OR

- **Speaker**: Modelling palaeoecological community data: a state-space approach

ResBaz - Research Bazaar *2021 – 2025*
Auckland, NZ; Madison, WI

- Organiser: ResBaz Aotearoa 2021.
- Workshop lead: Quarto for collaboration at ResBaz NZ 2022–2024 and in-person Madison, WI, 2023.

AGU - American Geophysical Union *2022*
Chicago, IL

- Speaker: Virtual ecological methods for assessing uncertainty in palaeoecological data using pseudo-proxies.

AMQUA - American Quaternary Association *2022*
Madison, WI

- Poster: Virtual ecological methods for assessing uncertainty in palaeoecological data using pseudo-proxies.

BES Palaeo in R - British Ecological Society *2022*
Virtual

- Speaker: Virtual ecological methods for assessing uncertainty in palaeoecological data using pseudo-proxies.

INQUA 2019 - International Union for Quaternary Research *2019*
Dublin, Ireland

- Speaker: Pseudoproxy modelling for assessing statistical inferences.

Durham University - Invited Speaker *2019*
Durham, UK

- Modelling resilience in ecosystems.

Biological Heritage Science Challenge *Christchurch, NZ*
University of Canterbury *2017 – 2019*

- Speaker on resilience in ecosystems.

International Swiss Climate Summer School *Bern, Switzerland*
University of Bern *2018*

- Poster presentation and participant. Accredited course (3 ECTS credits).

EcoRe3: Resistance Recovery and Resilience in Long-term Ecological Systems *Utah, USA*
PAGES - University of Utah *2018*

- Speaker: modelling resilience in ecosystems.

PEER REVIEW

Ecology Letters *2026*
Journal of Open Source Software *2026*
Nature Communications *2025*
Environmental Research and Technology *2025*
Quaternary Geochronology *2024*
Methods in Ecology and Evolution *2022*

SERVICE & LEADERSHIP

Early Career Representative *Virtual*
Neotoma Leadership Council *2024 – Present*

- Contributing to database governance, ingestion standards, and community user support for a global palaeoecological data resource.

Committee member *Virtual*
Computational Resources Committee *2025 – Present*

- Contributing to data infrastructure design and decisions, including security, data management, and FAIR/CARE practices.

Committee member *Virtual*
Postdoc, Research Associate, and Data Analyst/Data Scientist (PRADAS) committee *2026 – Present*

- Sharing information, training, and skills across the Cary PRADAS community